

Polished Questions and Solutions

1. If the sum of two numbers is 30 and the sum of their squares is 458, what is the square of their difference?

Solution: 16.

2. In how many ways can three couples be seated around a circular table such that no two adjacent persons are of the same gender and no spouse sits next to the other?

Solution: 2 ways.

3. Two men are 100 km apart. A moves toward B at 20 km/h, while B moves away from A at 10 km/h. A bee starting on A flies at 50 km/h back and forth between them until they meet. What distance does the bee cover?

Solution: 500 km.

4. A virus doubles every minute. If a jar gets completely filled in 12 minutes, how long does it take to fill one quarter of the jar?

Solution: 10 minutes.

5. A person travels from A to B at 8 km/h and from B to A at 2 km/h. What is the average speed for the round trip?

Solution: 3.2 km/h.

6. A cube has six faces and six distinct colors. How many unique colorings are possible, considering rotations?

Solution: 30.

7. In a right triangle ABC with $\angle A = 90^\circ$, the hypotenuse $BC = 25$ and the area is 84. What is the perimeter?

Solution: 56.

8. How many distinct permutations can be formed from the letters of the word "COCOON"?

Solution: 60.

9. Two partners invest in the ratio 4:5. They donate 10% of the total profit to charity. If A's share after charity is ₹10,00,000, what is the total profit?

Solution: ₹25,00,000.

10. A system has a 22-bit logical address space and a page size of 4 KB. How many entries are in the page table?

Solution: 1024 entries.

11. What happens when the following C code executes?

```
char *buff = "Nvidia";  
  
buff[0] = 'n';
```

Solution: It attempts to modify read-only memory (string literal) and causes undefined behavior.

12. Three processes with burst times 2, 4, and 8 arrive at time 0. Using Longest Remaining Time First (with lowest ID on ties), what is the average turnaround time?

Solution: 13.

13. When a queue is implemented using a singly linked list with a tail pointer, what is the time complexity of inserting (enqueue)?

Solution: $O(1)$.

14. What is the worst-case number of copy operations required to insert an element into a sorted array of length N ?

Solution: $N + 1$.

15. What operation is computed by this code?

```
k = 0;  
  
while (a - k*b >= b) ++k;  
  
return a - k*b;
```

Solution: It computes $a \% b$ (the modulo).

16. How can you determine whether a binary string represents a power of 2?

Solution: A binary number is a power of 2 if it contains exactly one '1'.